SUPPLEMENTAL MATERIAL

Table S1. Data Collection Sources and Atrial Fibrillation Definitions.

Study ID	Study Acronym	Trial Registration	Source	Definition	Source of definition	Type of AF event
Bailey, 2010[28]		NCT00528879	ClinicalTrials.gov	Serious adverse event	MedDRA 13.0	Not recorded
Barnett, 2014[29]	EMPA-REG RENAL	NCT01164501	ClinicalTrials.gov	Serious adverse event	MedDRA 15.0	Not recorded
Bode, 2013[30]		NCT01106651	ClinicalTrials.gov	Serious adverse event	MedDRA 14.0/16.0	Not recorded
Cannon, 2020[31]	VERTIS-CV	NCT01986881	ClinicalTrials.gov	Serious adverse event	MedDRA 22.1	Not recorded
Cefalu, 2015[32]		NCT01031680	ClinicalTrials.gov	Serious adverse event	MedDRA 14.0	Not recorded
Danne, 2019[33]	inTandem1	NCT02384941	ClinicalTrials.gov	Serious adverse event	MedDRA 20.0	Not recorded
Ferinnini, 2010[34]		NCT00528372	ClinicalTrials.gov	Serious adverse event	MedDRA 13.0	Not recorded
Frias, 2017[35]	DURATION-8	NCT02229396	ClinicalTrials.gov	Serious adverse event	MedDRA 20.1	Not recorded
Garg, 2017[36]	inTandem3	NCT02531035	ClinicalTrials.gov	Serious adverse event	MedDRA 20.0	Not recorded
Grunberger, 2017[37]	VERTIS RENAL	NCT01986855	Supplementary Appendix	Serious adverse event:	MedDRA 19.0	Not recorded
Haering, 2015[38]	EMPA-REG EXTEND MONO	NCT01289990	ClinicalTrials.gov	Serious adverse events	MedDRA 16.0	Not recorded
Heerspink, 2020[39]	DAPA-CKD	NCT03036150	ClinicalTrials.gov	Serious adverse events	MedDRA	Not recorded

Kovacs, 2015[40]	EMPA-REG PIOTM	NCT01210001	ClinicalTrials.gov	Serious + Non-serious adverse events	MedDRA 15.0	Not recorded
Mathieu, 2015[42]		NCT01646320	ClinicalTrials.gov	Serious + Non-serious adverse events	MedDRA 17.1	Not recorded
McMurray, 2019[25]	DAPA-HF	NCT03036124	ClinicalTrials.gov	Serious adverse events	MedDRA 22.0	Not recorded
Neal, 2015[26,43]	CANVAS	NCT01032629	ClinicalTrials.gov	Serious adverse events	MedDRA 19.1	Not recorded
Neal, 2017[26]	CANVAS-R	NCT01989754	ClinicalTrials.gov	Serious adverse events	MedDRA 19.1	Not recorded
Leiter, 2014		NCT01042977	ClinicalTrials.gov	Serious adverse events	MedDRA 14.0	Not recorded
Perkovic, 2019[44]	CREDENCE	NCT02065791	ClinicalTrials.gov	Serious adverse events	MedDRA 23.0	Not recorded
Wilding, 2013[45]	CANTATA-MSU	NCT01106625	ClinicalTrials.gov	Serious adverse events	MedDRA 21.0	Not recorded
Packer et al.[54]	EMPEROR-Reduced	NCT03057977	ClinicalTrials.gov	Serious adverse events	MedDRA 14.0/15.0	Not recorded
Pollock, 2019[46]	DELIGHT	NCT02547935	ClinicalTrials.gov	Serious adverse events	MedDRA 21.0	Not recorded
Pratley, 2018[47]	VERTIS-FACTORIAL	NCT02099110	ClinicalTrials.gov	Serious adverse events	MedDRA 19.0	Not recorded
Roden, 2013[48]		NCT01177813	ClinicalTrials.gov	Serious adverse events	MedDRA 14.1	Not recorded
Rosenstock, 2015[49]	EMPA-REG BASAL	NCT01011868	ClinicalTrials.gov	Serious adverse events	MedDRA 15.0	Not recorded

Rosenstock, 2018[50]	VERTIS-MET	NCT02033889	ClinicalTrials.gov	Serious adverse events	MedDRA 20.0	Not recorded
Softeland, 2017[51]		NCT01734785	Published paper	Serious + Non-serious adverse events	MedDRA 17.1	Not recorded
Wilding, 2012[52]		NCT00673231	ClinicalTrials.gov	Serious adverse events	MedDRA 12.0	Not recorded
Wiviot, 2019[24]	DECLARE-TIMI 58	NCT01730534	ClinicalTrials.gov	Serious adverse events	MedDRA 21.0	First event, Recurrent event, Hospitalization-associated events, Myocardial infarction-associated events and Heart Failure Hospitalization-associated events recorded
Yale, 2014[53]	DIA3004	NCT01064414	ClinicalTrials.gov	Serious adverse events	MedDRA 14.1/15.0	Not recorded
Zinman, 2015[55]	EMPA-REG OUTCOME	NCT01131676	ClinicalTrials.gov	Serious adverse events	MedDRA 18.0	Not recorded

MedDRA, Medical Dictionary for Regulatory Activities.

Table S2. Risk of Bias Assessment of Included Randomized Controlled Trials.

Study ID	Study Acronym	Trial Registration	Random Sequence Generation (Allocation Bias)	Allocation Concealment (Selection Bias)	Blinding of Participants and Researchers (Performance Bias)	Blinding of Outcome Assessment (Detection Bias)	Incomplete Outcome Data (Attrition Bias)	Other Bias
Bailey, 2010[28]		NCT00528879	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Barnett, 2014[29]	EMPA-REG RENAL	NCT01164501	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Bode, 2013[30]		NCT01106651	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Cannon, 2020[31]	VERTIS-CV	NCT01986881	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Cefalu, 2015[32]		NCT01031680	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Danne, 2019[33]	inTandem1	NCT02384941	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low
Ferinnini, 2010[34]		NCT00528372	Likely Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low

Frias, 2017[35]	DURATION-8	NCT02229396	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Garg, 2017[36]	inTandem3	NCT02531035	Likely Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Grunberger, 2017[37]	VERTIS RENAL	NCT01986855	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Haering, 2015[38]	EMPA-REG EXTEND MONO	NCT01289990	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Heerspink, 2020[39]	DAPA-CKD	NCT03036150	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Kovacs, 2015[40]	EMPA-REG PIOTM	NCT01210001	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low
Mathieu, 2015[42]		NCT01646320	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low
McMurray, 2019[25]	DAPA-HF	NCT03036124	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Neal, 2015[26,43]	CANVAS	NCT01032629	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low

Neal, 2017[26]	CANVAS-R	NCT01989754	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Leiter, 2014		NCT01042977	Likely Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Perkovic, 2019[44]	CREDENCE	NCT02065791	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Wilding, 2013[45]	CANTATA-MSU	NCT01106625	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Packer et al.[54]	EMPEROR-Reduced	NCT03057977	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low
Pollock, 2019[46]	DELIGHT	NCT02547935	Likely Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Pratley, 2018[47]	VERTIS-FACTORIAL	NCT02099110	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Roden, 2013[48]		NCT01177813	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low
Rosenstock, 2015[49]	EMPA-REG BASAL	NCT01011868	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low

Rosenstock, 2018[50]	VERTIS-MET	NCT02033889	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Softeland, 2017[51]		NCT01734785	Low	Likely Low	Low	AF/AFL Outcome: Likely high	Low	Low
Wilding, 2012[52]		NCT00673231	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Wiviot, 2019[24]	DECLARE-TIMI 58	NCT01730534	Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Yale, 2014[53]	DIA3004	NCT01064414	Likely Low	Low	Low	AF/AFL Outcome: Likely high	Low	Low
Zinman, 2015[55]	EMPA-REG OUTCOME	NCT01131676	Low	Low	Low	AF/AFL Outcome: Likely high HFH/CV Death: Low	Low	Low
Bhatt, 2020[56]	SOLOIST-WHF	NCT03521934	Low	Low	Low	HFH/CV Death: Low	Low	Low
Bhatt, 2020[57]	SCORED	NCT03315143	Low	Low	Low	HFH/CV Death: Low	Low	Low

AF, Atrial Fibrillation; AFl, Atrial Flutter; HFH, Heart Failure Hospitalization; CV Death, Cardiovascular Death.

Table S3. GRADE table, summarizing the evaluation of the quality of evidence.

Certainty assessment							№ of patients		Effect		Certainty
№ of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**[intervention]**	**[comparison]**	**Relative (95% CI)**	**Absolute (95% CI)**	
Serious AF/AFlutter Outcome											
31	randomised trials	serious	not serious	not serious	not serious	none	458/42931 (1.1%)	493/32348 (1.5%)	**RR 0.75** (0.66 to 0.86)	**4 fewer per 1,000** (from 5 fewer to 2 fewer)	⨁⨁⨁◯ MODERATE

CI: Confidence interval; **RR:** Risk ratio; **HR:** Hazard Ratio

Table S4. Pooled Odds ratio calculated using logistic regression with Firth's method treating study as fixed effect and fixed-effects meta-analysis without homogeneity assumption.

Methods	Odds ratio (95% CI)	P value
Logistic regression with Firth's method	0.75 (0.66-0.86)	<0.001
Fixed-effects meta-analysis without assuming homogeneity		
CMH estimator	0.75 (0.64-0.87)	<0.001
Woolf's estimator	0.75 (0.66-0.85)	<0.001
MLE	0.75 (0.66-0.86)	<0.001

CI: Confidence Interval; MLE: Maximum Likelihood Estimator; CMH: Cochran-Mantel-Haenszel

Figure S1. PRISMA Systematic Review Flow Diagram.

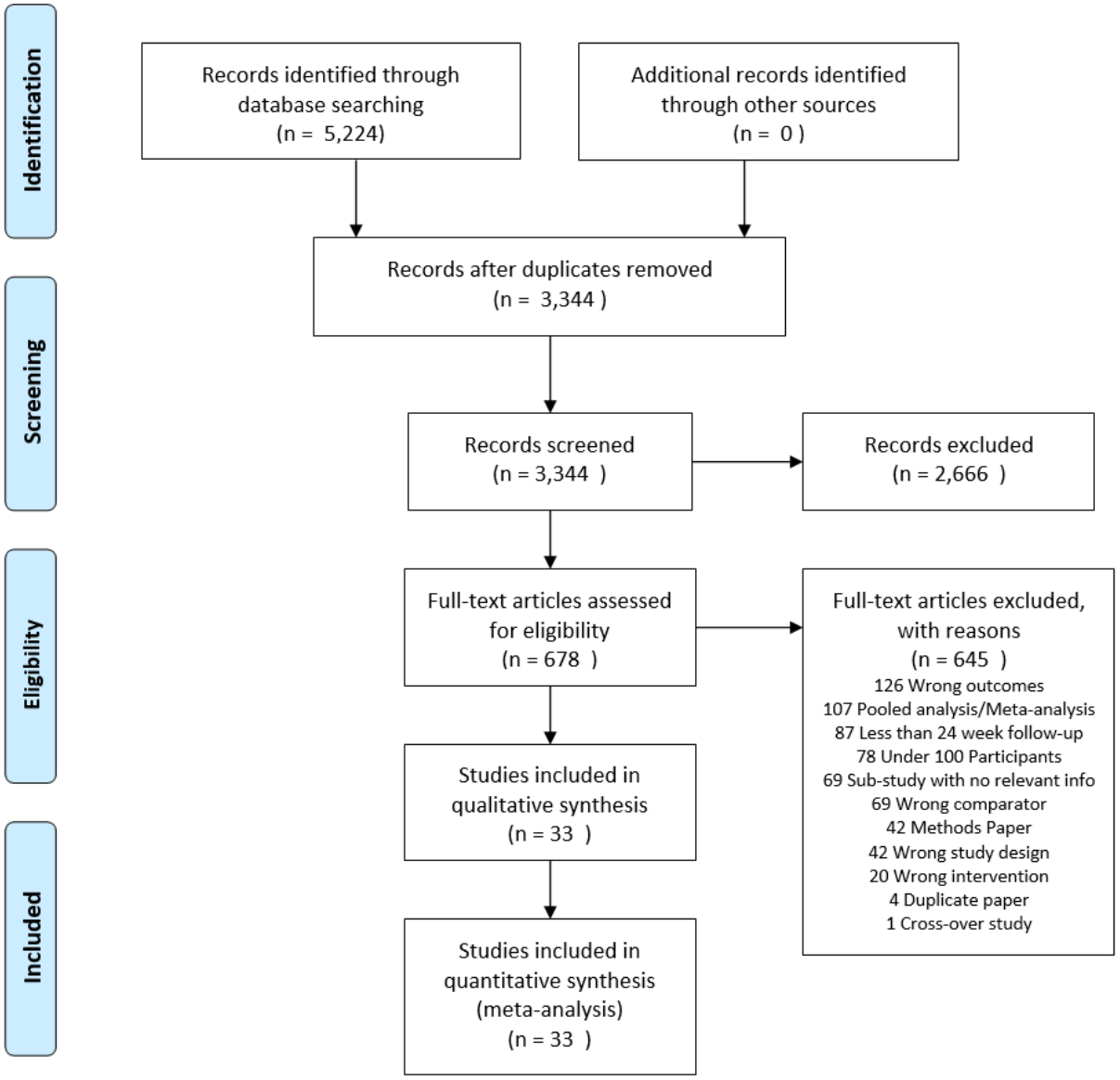
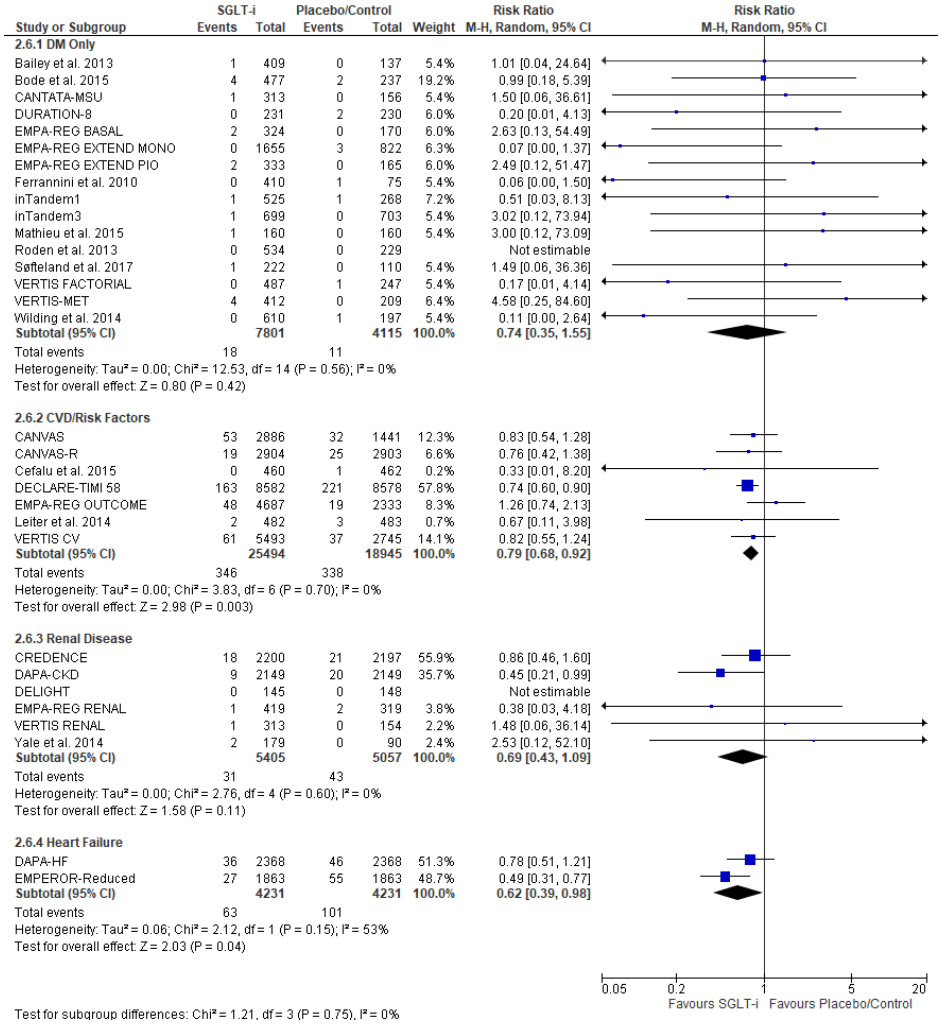


Figure S2. Forest plot demonstrating serious atrial fibrillation or atrial flutter events between patients on SGLT inhibitors versus placebo/control in Randomized Controlled Trials stratified by trial population.



Square markers represent point estimate of RR for individual studies, with square size proportional weight given to each study in the meta-analysis. Horizontal lines indicate 95% CIs. The solid diamond represents the estimated 95% CI for effect size of all meta-analysed data. CI, confidence interval; CVD, Cardiovascular Disease; DM, Diabetes Mellitus; RR, relative risk.

Figure S3. Funnel plot showing symmetrical distribution of studies indicating absence of publication bias.

